Assignment 2

Q1 (a)

$$\overline{(\bar{A}B + \bar{B}C)} + \overline{(D + \bar{B}C)}$$

$$= \overline{\bar{A}B + \bar{B}C} \cdot \overline{D + \bar{B}C}$$

$$= \overline{\bar{A}B} \cdot \overline{\bar{B}C} \cdot (D + \bar{B}C)$$

$$= (\bar{\bar{A}} + \bar{B})(\bar{\bar{B}} + \bar{C})(D + \bar{B}C)$$

$$= (A + \bar{B})(B + \bar{C})(D + \bar{B}C)$$

$$= (A + \bar{B})(BD + B\bar{B}C + \bar{C}D + \bar{B}C\bar{C})$$

$$= (A + \bar{B})(BD + 0 + \bar{C}D + 0)$$

$$= ABD + A\bar{C}D + \bar{B}BD + \bar{B}\bar{C}D$$

$$= ABD + A\bar{C}D + \bar{B}\bar{C}D$$

Q1 (b)

$$f(x, y, z) = \Pi(2, 6)$$

$$= \Pi(010, 110)$$

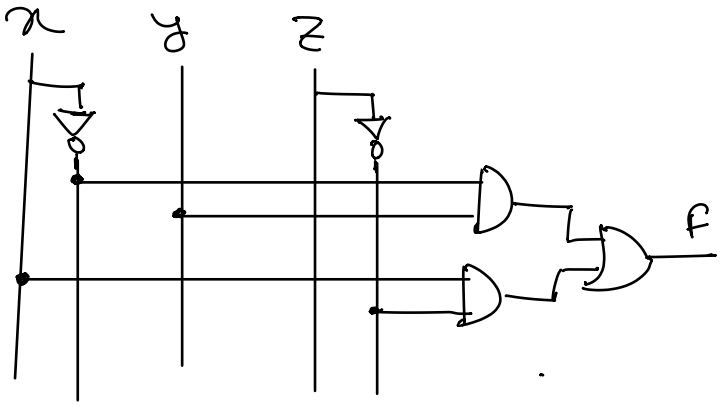
$$= (x + y' + z)(x' + y' + z)$$

$$= (y' + z + xx')$$

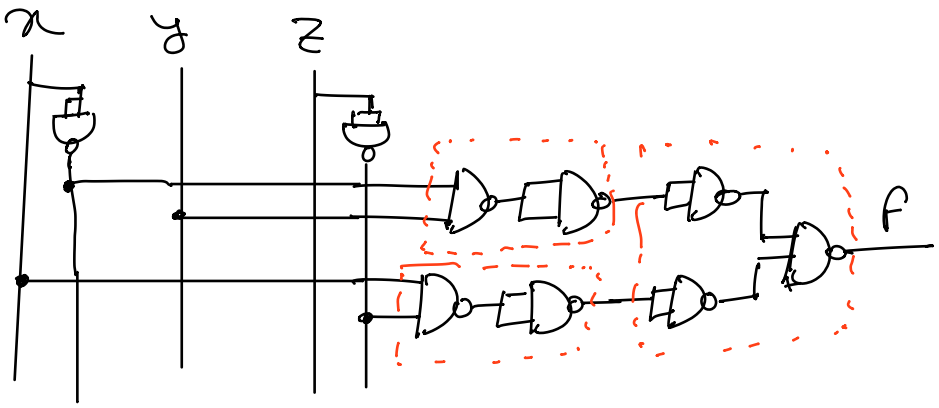
$$= y' + z$$

Ques 2

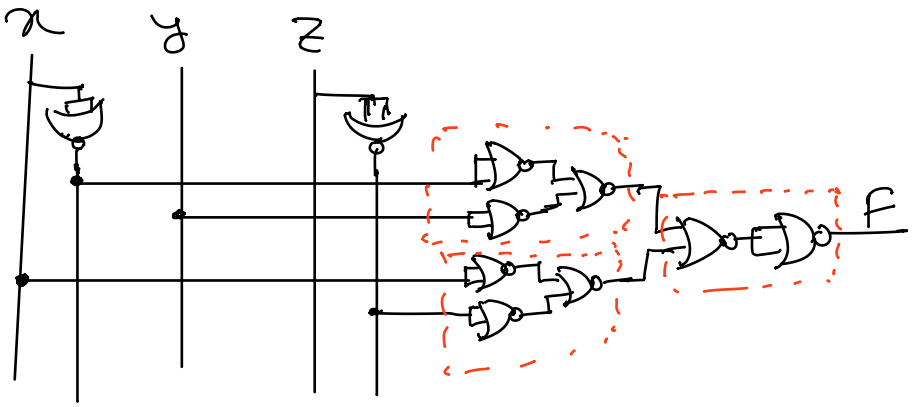
(a) $x'y + xz'$



Using NAND



Using NOR



Ques 3

(a) $F(a,b,c)$

$= b + b'c(a+c')$

$= b + ab'c + b'cc'$

$= b + ab'c + 0$

$= (a+b)(b+b')(b+c)$

$= (a+b)(b+c)$

$= (a+b+cc')(aa'+b+c)$

$= (a+b+c)(a+b+c')(a+b+c)(a'+b+c)$

$= \Pi(000, 001, 100)$

$= \Pi(0,1,4)$

(b) $F(w,x,y,z)$

$= x'y + y'(z+x'z')$

$= x'y + y'z + x'y'z'$

$= ((x'y+y')(x'y+z)) + x'y'z'$

$= ((x'+y')(y+y')(x'+z)(y+z)) + x'y'z'$

$= ((x'+y')(x'+z)(y+z)+x').((x'+y')(x'+z)(y+z)+y').$
$((x'+y')(x'+z)(y+z)+z')$

$= (x'+y'+x')(x'+z+x')(y+z+x')$
$(x'+y'+y')(x'+z+y')(y+z+y')$
$(x'+y'+z')(x'+z+z')(y+z+z')$

$= (x'+y')\ (x'+z)\ (x'+y+z)\ (x'+y')\ (x'+y'+z)$
$(x'+y'+z')$

$= (x'+y'+zz')\ (x'+yy'+z)\ (x'+y+z)\ (x'+y'+z)$
$(x'+y'+z')$

$= (x'+y'+z)\ (x'+y'+z')\ (x'+y+z)\ (x'+y'+z)$
$(x'+y+z)\ (x'+y'+z)\ (x'+y+z')$

$= (ww'+x'+y'+z)\ (ww'+x'+y'+z')\ +$
$(ww'+x'+y+z)\ (ww'+x+y+z')$

$= (w+x'+y'+z)\ (w'+x'+y'+z)\ (w+x'+y'+z')$
$(w'+x'+y'+z')\ (w+x'+y+z)\ (w'+x'+y+z)$

$= \Pi(0110, 1110, 0111, 1111, 0100, 1100)$

$= \Pi(6, 14, 7, 15, 4, 12)$

$= \Pi(4, 6, 7, 12, 14, 15)$

Ques 4 (b)

$((a+b+c')(a+b)(b'+c'))'$

$= (a+b+c')' + (a+b)' + (b'+c')'$

$= a'b'c + a'b' + bc$

$= a'b'(c+1) + bc$

$= a'b' + bc$

Ques 5

$f(P, Q, R, S)$

$= P + Q'RS'$

$= P(Q+Q') + Q'RS'$

$= PQ + PQ' + Q'RS'$

$= PQ(R+R') + PQ'(R+R') + Q'RS'$

$= PQR + PQR' + PQ'R + PQ'R' + Q'RS'$

$= PQR(S+S') + PQR'(S+S') + PQ'R(S+S') + PQ'R'(S+S')$
$+ (P+P')Q'RS'$

$= PQRS + PQRS' + PQR'S + PQR'S' + PQ'RS + PQ'RS'$
$+ PQ'R'S + PQ'R'S' + PQ'RS' + P'Q'RS'$

$= \Sigma(1111, 1110, 1101, 1100, 1011, 1010, 1001,$
$1000, 0010)$

$= \Sigma(15, 14, 13, 12, 11, 10, 9, 8, 2)$

$= \Sigma(2, 8, 9, 10, 11, 12, 13, 14, 15)$